

A4. Kompyuterni tashkil qilishning arifmetik asoslari. Turli xil sanoq tizimlarida arifmetik operatsiyalarni bajarish

Maqsad:

Talabalarga kompyuter arifmetikasining asosiy tushunchalarini o'rgatish, turli sanoq tizimlarida sonlar ustida arifmetik amallarni bajarish ko'nikmasini shakllantirish.

Vazifalar:

1. Sanoq tizimi tushunchasini o'rganish
2. Ikkilik, sakkizlik va o'n oltilik sanoq tizimlari bilan tanishish
3. Turli sanoq tizimlarida qo'shish, ayirish, ko'paytirish va bo'lishni bajarish
4. Sonlarni bir sanoq tizimidan boshqasiga o'tkazish

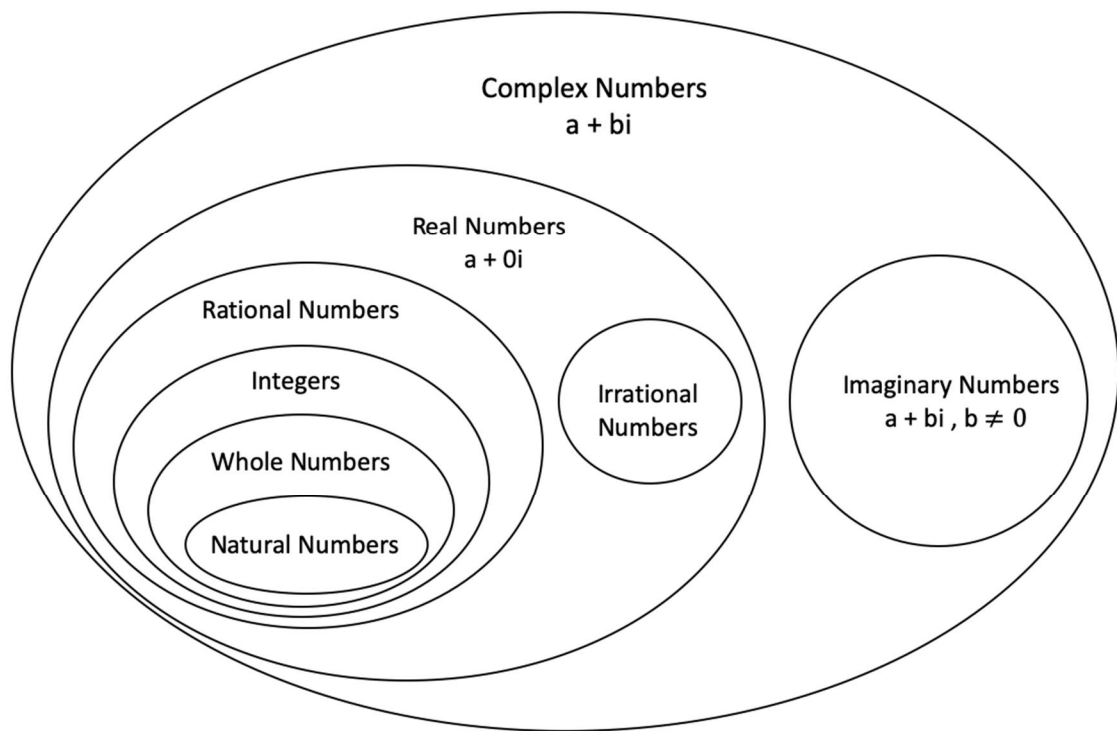
Nazariy qism

1. Sanoq tizimi tushunchasi

Sanoq tizimi — bu sonlarni yozish va ifodalash qoidalari majmui. Kompyuter texnikasida ma'lumotlar asosan **ikkilik sanoq tizimi** asosida qayta ishlanadi.

Sanoq tizimi Asosi Ishlatiladigan raqamlar

O'nlik	10	0–9
Ikkilik	2	0, 1
Sakkizlik	8	0–7
O'n oltilik	16	0–9, A–F



Decimal	Binary	Hexadecimal
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

2. Sonlarni sanoq tizimlari orasida o'tkazish

O'nlikdan ikkilikka o'tkazish (bo'lish usuli)

Masalan: 13_{10}

13	÷	2	=	6	(qoldiq	1)
6	÷	2	=	3	(qoldiq	0)
3	÷	2	=	1	(qoldiq	1)
$1 \div 2 = 0$ (qoldiq 1)						

👉 Natija: $13_{10} = 1101_2$

Decimal to Binary Conversion



Step 1: Divide the given number **13** repeatedly by 2 until you get '0' as the quotient

$13 \div 2 = 6$	(Remainder 1)	
$6 \div 2 = 3$	(Remainder 0)	
$3 \div 2 = 1$	(Remainder 1)	
$1 \div 2 = 0$	(Remainder 1)	

Step 2: Write the remainders in the reverse order **1 1 0 1**

$$\therefore 13_{10} = 1101_2$$

(Decimal) (Binary)

Remainder:

0
0
1
1
1
0
0
1

$156_{10} = 10011100_2$

wiki How to Convert from Decimal to Binary

3. Ikkilik sanoq tizimida arifmetik amallar

a) Ikkilik qo'shish

$$\begin{array}{r}
 1011 \\
 + 0110 \\
 \hline
 10001
 \end{array}$$

Qoida:

- $0 + 0 = 0$
- $0 + 1 = 1$
- $1 + 1 = 10$ (0 yoziladi, 1 o'tadi)

What is Binary Addition?

Binary Addition Example

carry into the second column

0	10	110	0110
0110	0110	0110	0110
0111	0111	0111	0111
1	01	101	1101
result for first column			

final result

A + B	S
0 + 0	
0 + 1	
1 + 0	
1 + 1	

Binary Addition Example

carry into the second column

0	10	110	0110
0110	0110	0110	0110
0111	0111	0111	0111
1	01	101	1101
result for first column			

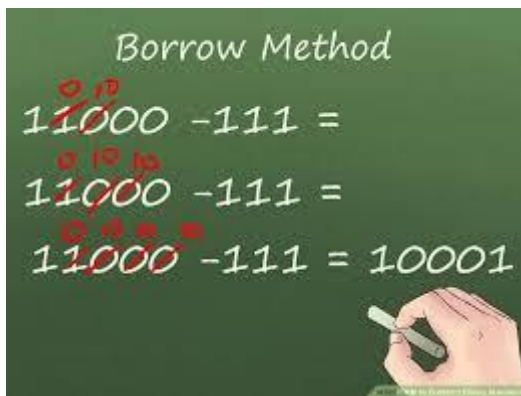
final result

b) Ikkilik ayirish

$$\begin{array}{r}
 1010 \\
 - 0011 \\
 \hline
 0111
 \end{array}$$

Qoida:

- 0 dan 1 ayirilsa, **qarz olinadi**



c) Ikkilik ko‘paytirish

$$\begin{array}{r} 101 \\ \times 11 \\ \hline 101 \\ 101 \\ \hline 1111 \end{array}$$

$$\begin{array}{r} 11011 \\ \times 101 \\ \hline 11011 \\ 110110 \\ 1101100 \\ \hline 10000111 \end{array}$$

$$\begin{array}{r}
 101101 \\
 1101 \\
 \hline
 101101 \\
 000000 \\
 \hline
 0101101 \text{ First Intermediate sum} \\
 101101 \\
 \hline
 11100001 \text{ Second Intermediate Sum} \\
 101101 \\
 \hline
 1001001001 \text{ Final Sum}
 \end{array}$$

d) Ikkilik bo'lish

Ikkilik bo'lish o'nlik bo'lishga o'xshash, faqat 0 va 1 ishlatiladi.

$$\begin{array}{r}
 10100.1110 \\
 \hline
 100 \overline{) 1010011.1010} \\
 \underline{- 100} \\
 100 \\
 \underline{- 100} \\
 111 \\
 \underline{- 100} \\
 110 \\
 \underline{- 100} \\
 101 \\
 \underline{- 100} \\
 10
 \end{array}$$

$$(1010011.1010)_2 \div (100)_2 = (10100.110)_2$$

Binary Division: Example

$$\begin{array}{r} 101 \\ 101 \overline{) 11010} \\ \underline{(-) 101} \downarrow \\ 11 \downarrow \\ \underline{(-) 00} \downarrow \\ 110 \downarrow \\ \underline{(-) 101} \\ 1 \end{array}$$

4. O‘n oltilik sanoq tizimi (Hexadecimal)

O‘nlik Hex

10	A
11	B
12	C
13	D
14	E
15	F

Masalan:

$$2F_{16} = 0010\ 1111_2$$

Decimal	Hexadecimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

Decimal Numbers	4-bit Binary Number	Hexadecimal Number
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
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12	1100	C
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Amaliy topshiriqlar

- Quyidagi sonlarni ikkilik sanoq tizimiga o'tkazing:
 - 25_{10}
 - 42_{10}
- Ikkilik sanoq tizimida hisoblang:
 - $10101_2 + 1101_2$
 - $10010_2 - 1011_2$
- $3A_{16}$ sonini o'nlik va ikkilik sanoq tizimlariga o'tkazing.
- Quyidagi amalni bajaring:
 $1110_2 \times 101_2$

Xulosa

Ushbu amaliy ish orqali talabalar kompyuter arifmetikasining asoslarini, turli sanoq tizimlarida ishlashni va arifmetik amallarni bajarishni o'rganadilar. Bu bilimlar protsessor ishlashi, dasturlash va raqamli texnika fanlari uchun muhim hisoblanadi.

Nazorat savollari

1. Sanoq tizimi nima?
2. Ikkilik sanoq tizimining asosi nechaga teng?
3. O'n oltilik sanoq tizimi qayerlarda qo'llaniladi?
4. Ikkilik qo'shish qanday bajariladi?
5. Nima sababdan kompyuterlar ikkilik sanoq tizimida ishlaydi?